Project:
Mental Health Assisstant
(Depression Detection Using DistilBERT)

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1. Project Overview

This project presents a Mental Health Assistant designed to detect depression indicators in social media text. By fine-tuning the DistilBERT transformer model on a large multi-source dataset, the system classifies user-generated text as either depressive or non-depressive with high accuracy. The ultimate goal is to serve as an intelligent backbone for a conversational therapy assistant capable of identifying emotional distress in real time.

1.1 Objectives

- Build a robust binary classification model for depression detection from social media text.
- Perform thorough exploratory data analysis to uncover linguistic patterns associated with mental distress.
- Fine-tune DistilBERT on combined Reddit and Twitter datasets for optimal performance.
- Extend the system into a conversational AI therapy assistant using Ollama Llama 3.2 3B.

1.2 Workflow Summary

The project follows a structured ML pipeline:

(1) multi-source data ingestion

(2) data quality checks and cleaning

(3) exploratory data analysis

(4) text preprocessing and tokenization

(5) DistilBERT fine-tuning with early stopping

(6) evaluation on a held-out test set and an external dataset

(7) integration with a conversational LLM for deployment as a therapy assistant.

2. Data Summary and Description

Three publicly available datasets were sourced from Kaggle and combined to create a rich, diverse corpus for model training. The datasets span two major social media platforms Reddit and Twitter ensuring the model learns from varied writing styles, vocabularies, and expression patterns.

2.1 Source Datasets

Dataset	Source	Rows	Class Balance
Dataset 1	Reddit (infamouscoder) Kaggle link	7,731	Not Depression: 3,900 Depression: 3,831
Dataset 2	Twitter Train (samrats) Kaggle link	31,000	Not Depression: 15,000 Depression: 16,000
Dataset 3	Twitter Test (samrats) Kaggle link	31,000	Not Depression: 15,000 Depression: 16,000

2.2 Data Quality Overview

Dataset	Missing Values	Duplicate Rows
Reddit Dataset	None	81 rows
Twitter Train	None	932 rows
Twitter Test	None	1,956 rows

2.3 Combined Dataset

After concatenating the three datasets and standardising column names to 'text' and 'label', the combined dataset contained 69,731 rows. Following the removal of 2,969 duplicate records, the final cleaned dataset comprised 66,762 unique text samples with a near-balanced class distribution.

Metric	Value
Total Rows (raw)	69,731
Duplicate Rows Removed	2,969
Final Unique Rows	66,762
Not Depressed (Label 0)	~35,000 (~52%)
Depressed (Label 1)	~32,000 (~48%)

3. Data Cleaning Steps

A multi-step data cleaning pipeline was applied to ensure data quality and model readiness. Each step was deliberately designed to preserve emotional signal while removing noise.

3.1 Duplicate Removal

Exact duplicate rows were identified and removed. A total of 2,969 duplicate entries were found across the combined dataset. Removing duplicates prevents the model from over-fitting to repeated patterns and ensures each training sample contributes uniquely to learning.

3.2 Text Preprocessing for Modeling

A dedicated clean_text() function was applied to all text samples prior to training:

- URL Removal: All HTTP/HTTPS URLs were stripped using re.sub(r"http\S+", "", text) to eliminate irrelevant hyperlinks that carry no semantic value.
- HTML Tag Removal: Residual HTML markup was removed using re.sub(r"<.*?>", "", text) to clean any web-scraped artefacts.
- Whitespace Normalisation: Multiple consecutive spaces and tabs were collapsed to single spaces and leading/trailing whitespace was trimmed.

3.3 Train / Validation / Test Split

The cleaned dataset was split into three stratified subsets to ensure each partition maintains the same class distribution as the full dataset:

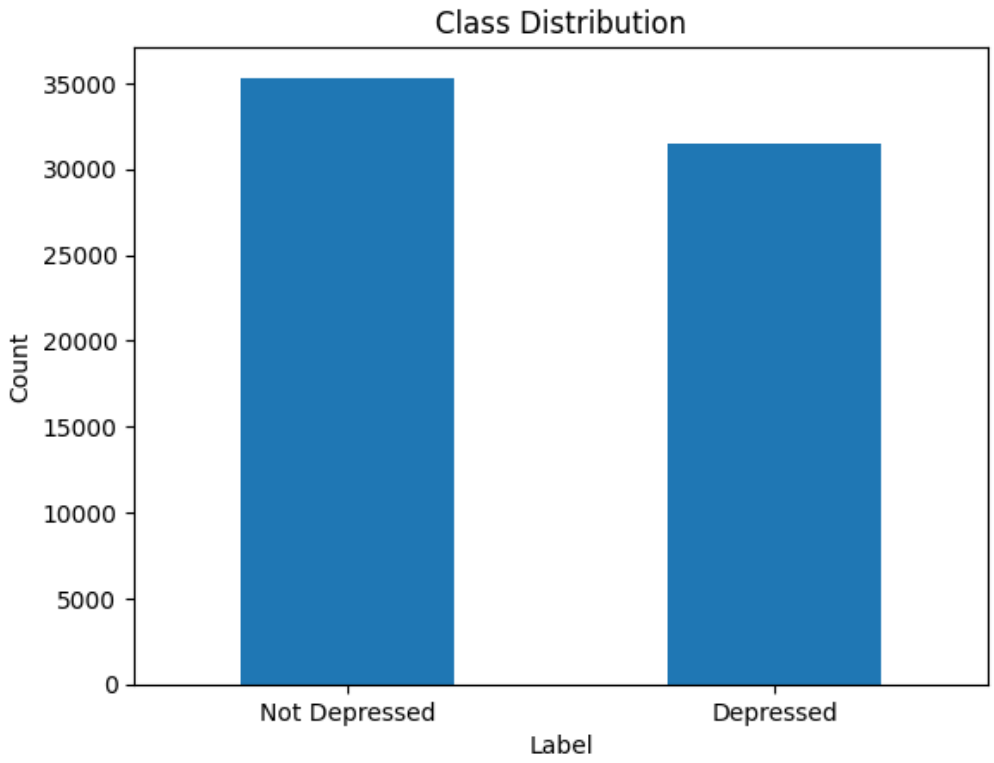
Split	Size	Proportion	Purpose
Training Set	46,733	70%	Model fine-tuning
Validation Set	10,014	15%	Hyperparameter tuning & early stopping
Test Set	10,015	15%	Final unbiased evaluation

4. Exploratory Data Analysis (EDA)

A comprehensive EDA was conducted to understand the dataset's structure, uncover linguistic patterns, and validate the suitability of the data for binary classification. The analysis covers class distribution, text length and word count, frequent word patterns, and special character usage.

4.1 Class Distribution

Plot: Bar Chart Class Distribution

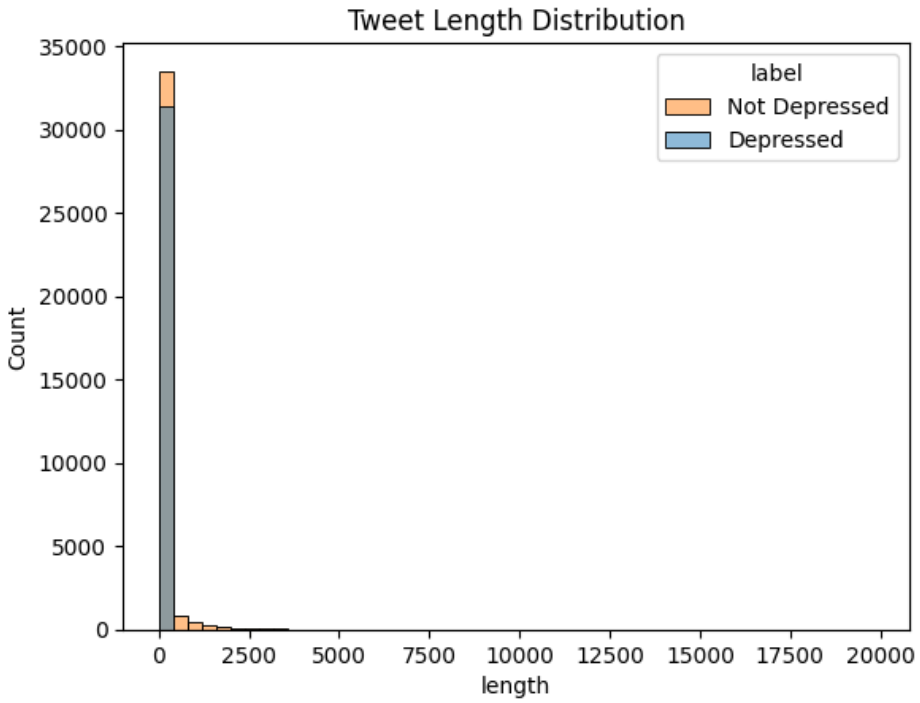


Insight:

- The dataset is well-balanced, with approximately 35,000 samples labelled as Not Depressed and 32,000 labelled as Depressed.
- The near-even class split (~52% vs 48%) is ideal for binary classification as it prevents the model from biasing towards a majority class.
- Because the classes are balanced, Accuracy is a reliable primary metric. In an imbalanced setting, F1-score or AUC would be preferred.

4.2 Text Length Distribution

Plot: Histogram Tweet / Post Length Distribution by Class

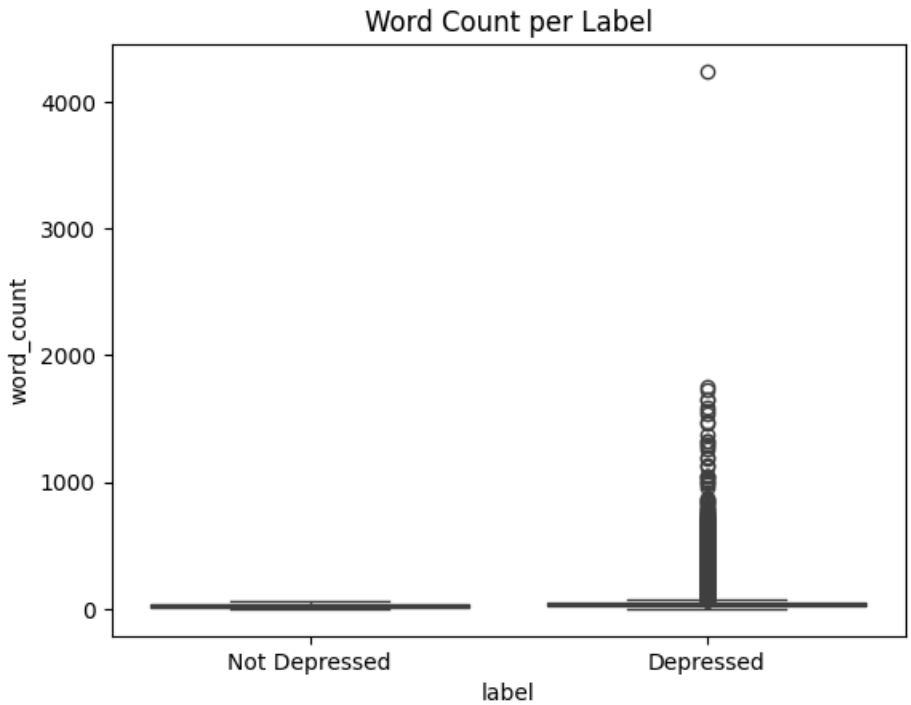


Insight:

- Both classes are heavily right-skewed, meaning the vast majority of posts are under 500 characters typical of social media data.
- The Depressed class shows a notable long tail, with some posts exceeding 10,000–20,000 characters, representing long-form venting or detailed personal narratives.
- Since DistilBERT has a 128-token maximum, long posts are truncated. The first 128 tokens are generally sufficient to capture the emotional tone.

4.3 Word Count per Label

Plot: Box Plot — Word Count per Labe

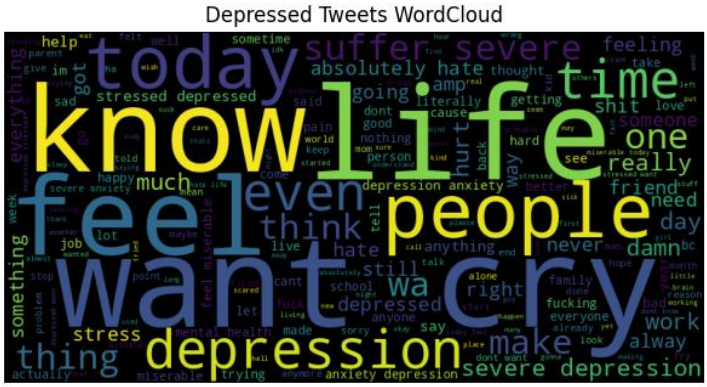


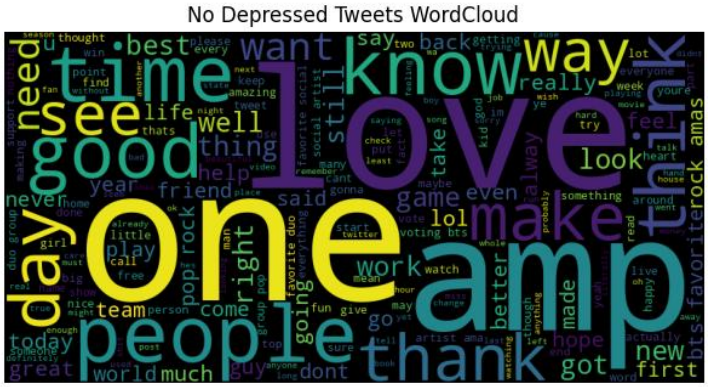
Insight:

- The median word count is low and similar for both classes, indicating that the average post in both categories is concise.
- The Depressed class shows a dramatically wider distribution of outliers, with some posts exceeding 4,000 words a pattern absent in the Not Depressed class.
- This confirms the 'venting phenomenon': individuals expressing depressive symptoms disproportionately use social media for extended personal storytelling.

4.4 Word Frequency Analysis & Word Clouds

Plot: Word Clouds Depressed vs. Not Depressed Classe




No Depressed Tweets WordCloud
love
one
amp
people
time
know
way
good
see
make
thank
want
day
need
work
look
game
lol
better
made
new
first
got
favorite
rock
amas
bts
hope
great
world
much
dont
come
team
play
right
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best
well
thing
friend
year
still
back
say
really
life
thats
amazing
take
feel
never
even
something
around
read
twitter
voting bts
girl
little
call
free
nice
enough
sure
artist
anyone
post
watching
though
anything
change
may
give
fun
go
going
today
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dont
watch
mean
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whole
week
everyone
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hard
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house
kid
god
jim
wish
movie
song
saying
check
put
least
remember
cant
gonna
maybe
start
twitter
favorite duo
favorite social
social artist
artist and
artist
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anything
change
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girl
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enough
sure
artist
anyone
post
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change
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give
fun
go
going
today
guy
dont
watch
mean
hour
whole
week
everyone
your
try
hard
talk
heart
house
kid
god
jim

5.2 Tokenisation

Texts were tokenised using the AutoTokenizer for distilbert-base-uncased with the following settings:

- Truncation: True long texts are truncated to the maximum length.
- Padding: max_length all sequences are padded to a uniform length of 128 tokens.
- Max Length: 128 tokens sufficient for the vast majority of social media posts.

5.3 Training Results per Epoch

The best model checkpoint was saved at Epoch 4 (highest F1 = 99.53%).

Epoch	Train Loss	Val Loss	Accuracy	F1	Precision	Recall
1	0.0852	0.0529	99.40%	99.43%	99.64%	99.23%
2	0.0422	0.0596	99.38%	99.41%	99.49%	99.34%
3	0.0173	0.0579	99.41%	99.44%	99.42%	99.47%
4	0.0035	0.0683	99.50%	99.53%	99.58%	99.47%
5	0.0010	0.0794	99.48%	99.51%	99.55%	99.47%

6. Evaluation Results

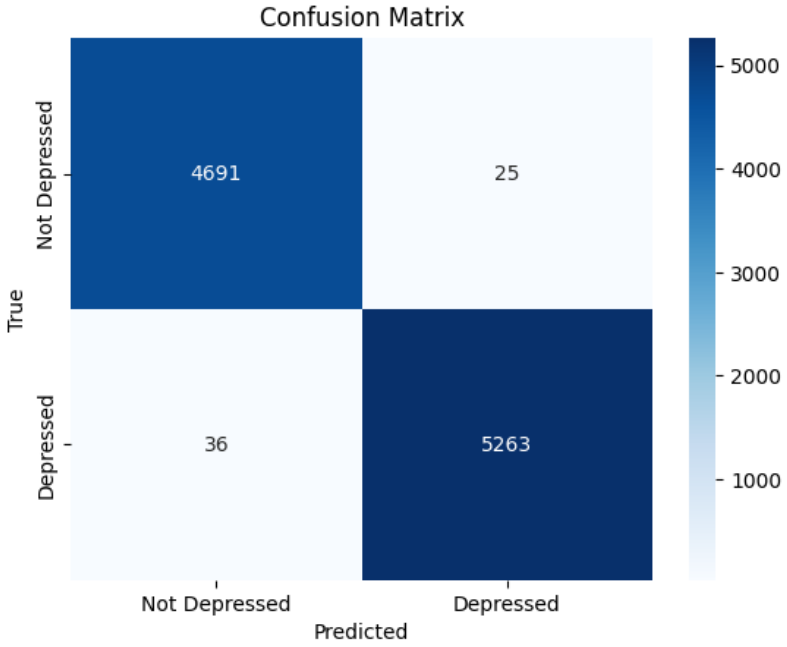
6.1 Internal Test Set Performance

The saved best model was evaluated on the 10,015 held-out test samples that were never seen during training or validation:

Metric	Not Depressed	Depressed	Overall
Precision	0.99	1.00	0.99
Recall	0.99	0.99	0.99
F1-Score	0.99	0.99	0.99
Support	4,716	5,299	10,015
Accuracy	—	—	99.39%

6.2 Confusion Matrix Internal Test Set

Plot: Confusion Matrix Internal Test Set



The confusion matrix shows only 25 false positives and 36 false negatives out of 10,015 predictions, demonstrating exceptional discrimination between the two classes.

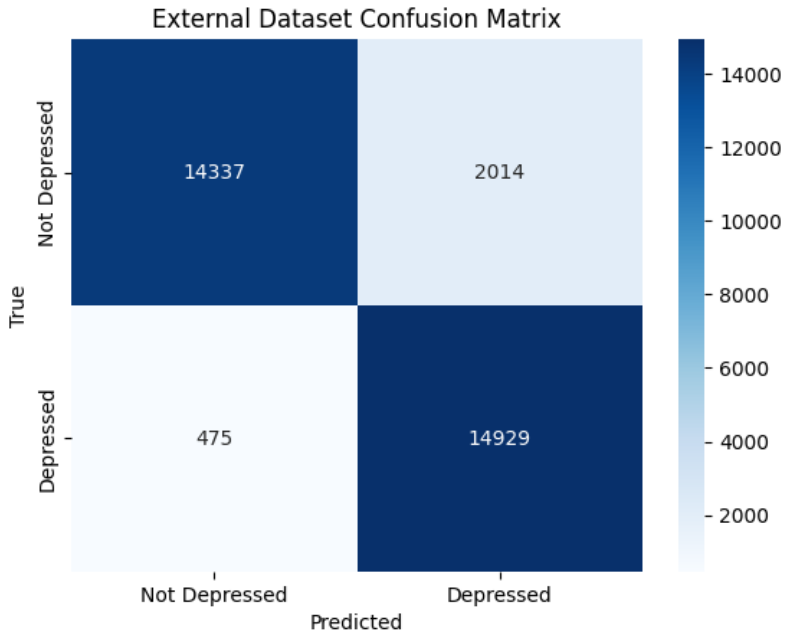
6.3 External Dataset Validation (Kaggle link)

To assess real-world generalisation, the model was evaluated on an independent external dataset (Suchintika Sarkar / Sentiment Analysis for Mental Health), which was not part of the original training pipeline:

Metric	Not Depressed (Normal)	Depressed (Depression)	Overall
Precision	0.97	0.88	0.92
Recall	0.88	0.97	0.92
F1-Score	0.92	0.92	0.92
Support	16,351	15,404	31,755
Accuracy	—	—	92.16%

6.4 Confusion Matrix External Dataset

Plot: Confusion Matrix External Dataset



Insight:

To verify the model's generalizability, it was evaluated on an external dataset. While accuracy dropped from 99% to 92.1%, the model maintained a high Recall of 0.97 for the Depression class. This indicates that the model is highly effective at identifying depressive signals across different data sources, though it tends toward 'over-sensitivity' (False Positives) when analyzing normal everyday stress.

7. Enhancement: Chatbot with Ollama Llama 3.2 3B

To transform the depression detection model into a fully interactive mental health assistant, the project is extended with a conversational AI layer powered by Ollama's locally-hosted Llama 3.2 3B model. This creates a two-stage pipeline: first classify the user's emotional state, then respond with contextually appropriate, empathetic dialogue.

7.1 System Architecture

The enhancement introduces a dual-component pipeline:

- Stage 1 Classifier: The fine-tuned DistilBERT model analyses the user's input and predicts a depression risk label (0 = Not Depressed, 1 = Depressed).
- Stage 2 Conversational LLM: Based on the classifier's output, a dynamically constructed system prompt is fed to the Llama 3.2 3B model via Ollama's local API, generating a therapeutic, empathetic response.